Predicting Plant Survival With Color Vision

1. Introduction

Exploratory Data Analysis of the pennycress data resulted in some discoveries about color vision values as a plant grows. Using these findings, my team and I decided to run a few models. The goal of these models was to determine if we can predict plant survival based on information from the color vision cameras. We are exploring this idea since it would be beneficial to researchers, as it will save them time. In the past, researchers would have to measure each individual plant multiple times to track growth. Color vision cameras can assist with this by recording those measurements automatically from the images taken. The model we are trying to find will save them more time since it will be able to tell them if a plant will live based on the first week or two of observations. Therefore, rather than spending a month on the same plants being monitored, they can use the model to get results in as little as two weeks.

2. Background

Catboost (also known as categorical boosting) is a decision tree based modeling technique. This technique handles categorical variables without having to one-hot encode. In one-hot encoding, a new column is created for each unique value of that variable, and it gets filled with 1's (if that value is true for the observation) and 0's (if it is false). Catboost also handles missing values well without having to fill in the data through imputation prior to running the model. Data imputation takes the most plausible value and fills in the missing data, but takes time. Considering the pennycress data had many missing values, this was a beneficial feature of Catboost. Additionally, Catboost works with large datasets well since it uses parallel processing to compute the decision trees quickly. Due to these features aligning with our data, Catboost became a useful modeling technique.

The technique is based on decision trees, which is where the data is split up by multiple decisions. It takes the full dataset, and asks a question, such as "Is the area above 20,000?". If the answer is yes, then that plant will be split into the yes group. If the answer was no, then the plant is split into the no group. Then it will take the yes group and ask it another question, and so on for every group and

question. These questions will lead to a prediction on whether or not the plant will live. Once that prediction has been made on a specific plant, it will check the result with the true result recorded. Multiple of these decision trees will be made, and it will take errors from the previous one to make the next one better. This continues until it reaches a certain number of iterations or hits a certain condition. To prevent overfitting, Catboost has an option to stop running after a set number of trees have similar accuracy. Once that number is reached, it will not use any of the iterations after the first instance of that accuracy.

3. Data preprocessing

The raw data provided included the mean red, green, and blue values for a specific observation. These are achieved by taking the average color value of all the pixels in the given image, excluding the background. Consequently, these values are highly sensitive to changes in shadows, lighting, and camera exposure. To get a clearer idea of the actual colors of the plant, we used established visible-spectrum vegetation indices. The raw values are converted into chromatic coordinates, or proportions, by dividing each color by the sum of all three. These coordinates were then used to calculate the indices. We used EXG (Excess Green) and NGRDI (Normalized Green Red Difference Index) which show chlorophyll concentration and overall biomass. We also calculated VARI (Visible Atmospherically Resistant Index), which was formulated to find the total fraction of living vegetation while accounting for atmospheric scattering and lighting differences. Finally, EXR (Excess Red) was calculated, which shows necrosis and environmental stress. All four of these indices are widely used in precision agriculture and agronomy, making them reliable methods to evaluate the true biological pigment changes rather than artificial lighting fluctuations.

A binary target variable for plant survival was also appended to the dataset. The plants were classified based on a maximum growth threshold of 20,000 pixels. This was calculated for each unique plant across the entire experiment. If a plant reached or exceeded this threshold at any point, it was classified as alive (encoded as 1). Conversely, plants that failed to achieve this maximum were classified as dead (encoded as 0).

To capture the early growth trends rather than daily fluctuations, the data was aggregated based on specific time windows. The mean EXG, EXR, VARI, and

NGRDI values were calculated for a 5, 7, 10, and 14-day window. These averages were used as predictors in the Catboost algorithm. The total number of plants for each of these windows varied slightly. This was likely due to plants being dropped from the data due to missing values. For example, in cases where a plant started growing late, it would show up in the 10 or 14 day windows, but would not appear in the 5 or 7. The discrepancy was accounted for by looking at proportions rather than exact numbers.

4. Modeling

To begin modeling, the dataset was partitioned into an 80% training set and a 20% testing set. Ensuring the ratio of living to dead plants remained consistent across both sets, stratified sampling was utilized. Due to the dataset being imbalanced with about a three to one ratio of living to dead plants, balanced class weights were applied during training. The balanced weights appropriately penalize minority class misclassifications. We began by training the CatBoost algorithm on the 5-day window features. An early stopping parameter was implemented, which stops iterations if the accuracy is similar for 50 consecutive iterations, to prevent overfitting. The model was used on the testing set to predict survival based on what the model learned. Finally, these predictions were compared against the true observed outcomes. Model performance was evaluated using overall accuracy and a confusion matrix, which outlines the exact distribution of true positive, true negative, false positive, and false negative classifications.

Initial modeling using the 5-day window averages yielded a 72% overall accuracy and a 71% minority class recall. In this agricultural context, maximizing recall for the dead plants is important; misclassifying a dying plant as a survivor wastes resources on an unsalvageable crop. To improve this sensitivity, the observation window was expanded. The 7-day model demonstrated significant improvement, achieving 79% accuracy and peaking at an 82% recall rate. While extending the windows to 10 and 14 days incrementally improved overall accuracy (80% and 84%, respectively), recall stagnated at 77% for both. Due to the early detection of fatal stress being the primary objective, the 7-day window was identified as the optimal temporal baseline.

Before finalizing the feature set, the rate of change (slope) for each index was evaluated to determine if daily trajectory was a stronger predictor than the averages. A model trained exclusively on 5-day slopes achieved 73% accuracy but

a poor 64% recall. Attempting to combine both the slopes and the averages introduced severe multicollinearity and statistical noise, which reduced the model's predictive capabilities. Consequently, trajectory metrics were discarded in favor of the cleaner, aggregated averages.

Finally, feature importance metrics were extracted to evaluate the predictive weight of each individual index. Across all multivariate models, EXG and the raw red pixel average consistently emerged as the most influential variables. To test the efficiency of a simplified algorithm, a new model utilizing only the 7-day EXG average was used. This EXG model achieved an 84% accuracy while maintaining the peak 82% recall rate—outperforming the 7-day model in overall accuracy without sacrificing sensitivity. To verify that this success was due to the predictive power of EXG values rather than simply single-metric modeling, models were created for EXR, VARI, and NGRDI. None of these alternative models exceeded a 68% accuracy nor a 76% recall. Therefore, the 7-day EXG-only classifier was selected as the overall best model.